

teaching. In India, students need to memorise their course material. We need to find ways of changing how to teach in order to help students understand their course instead of just memorising the material. Further, once you go beyond some of the top universities in India, the quality of education you get is very questionable. Most Tier-II and Tier-III universities do not have enough faculty to teach different subjects.

Universities in India should allow students to take some online courses, especially in the areas they do not have their own faculty. Local education institutions should also let their students take a full course through an online platform like edX, but appear for examinations under a local teaching assistant. We are already seeing some success in using edX to improve the Indian education system.

Teachers and professors can take courses about using blended learning on edX to learn how they can integrate these learning methods in their teaching practices. This will uplift the education system.

Q Countries like Jordan, China and Israel are already using Open edX to launch their national education platform. What about your plans for India?

We are in talks with certain sections of the Indian government and would be delighted to work for millions of Indians. We would love to see the government deploy Open edX and launch a national platform to spread the next generation of education in India.

The Tatas have already announced that they are set to launch a platform using Open edX for high-school students. The same platform can be used for skills, college education and teacher training.

Q How do Indian developers benefit from a platform like edX?

Many companies all over the world are building on Open edX and providing services to corporations and governments, for a certain fee. Entrepreneurs and software developers in India can utilise the same platform and start offering some new learning models. There is a vast ecosystem and many companies worldwide are utilising our software to garner profits.

Q Last of all, how can edX help the jobless skilled workers in India?

I would encourage all the jobless skilled workers to go and take courses on edX. We have courses to nurture many in-demand skills. These are designed by institutions like IIT Bombay, MIT or Berkeley. In fact, just this summer, we've launched some of the hottest job-related courses to reduce unemployment.

You don't need to spend hours sitting in a classroom to learn something new and interesting. Instead, all you need is an active Internet connection to enhance your skills using edX. 

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17. Distributional semantics is based on the idea that semantic similarities can be characterised based on the distributional properties of the words in a large corpus. This is popularly expressed by the statement, "A word is known by the company it keeps." If two words have similar distributions or similar contexts, then they have similar meanings. Can you give a couple of examples of the applications of distributional semantics in natural language processing?
18. Machine translation is the automatic translation of text from a source language to a target language. Machine translation can either be driven by rule based techniques or by statistical techniques. Can you explain the advantages and disadvantages of both rule based machine translation and statistical machine translation?
19. What is transfer learning? Can you illustrate it with an example?
20. Many of the standard NLP systems have been designed for reasonable document sizes, be it automatic key phrase recognition, topic modelling or summarisation. Consider

that you have been asked to design a NLP pipeline to analyse the tweet texts. Tweets are typically very short text and are characterised by the extensive use of non-standard abbreviations. What are the major NLP challenges in analysing short text?

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, happy programming! 

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